

Qatar-1b

R-0618

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_240831 · created 2026-07-25T11:14:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460553.8125 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.033
Transit depth (Rp/Rs)^2	1.22 +/- 0.73 [%]
Orbital Inclination (inc)	86.05 +/- 2.32
Airmass coefficient 1 (a1)	1.011 +/- 0.023
Airmass coefficient 2 (a2)	-0.008 +/- 0.018
Scatter in the residuals of the lightcurve fit is	2.31 %
Best Comparison Star	#1 - [495, 356]
Optimal Method	PSF photometry
Transit Duration (day)	0.072 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.111 ± 0.033 vs 0.1463 (deviation 0.71σ)
Duration fit vs published	0.072 d vs 0.0692 d (deviation 0.19σ)
Mid-time O-C	-7.2 min
Depth SNR	2.2
Residual scatter	2.31 %

Light curve

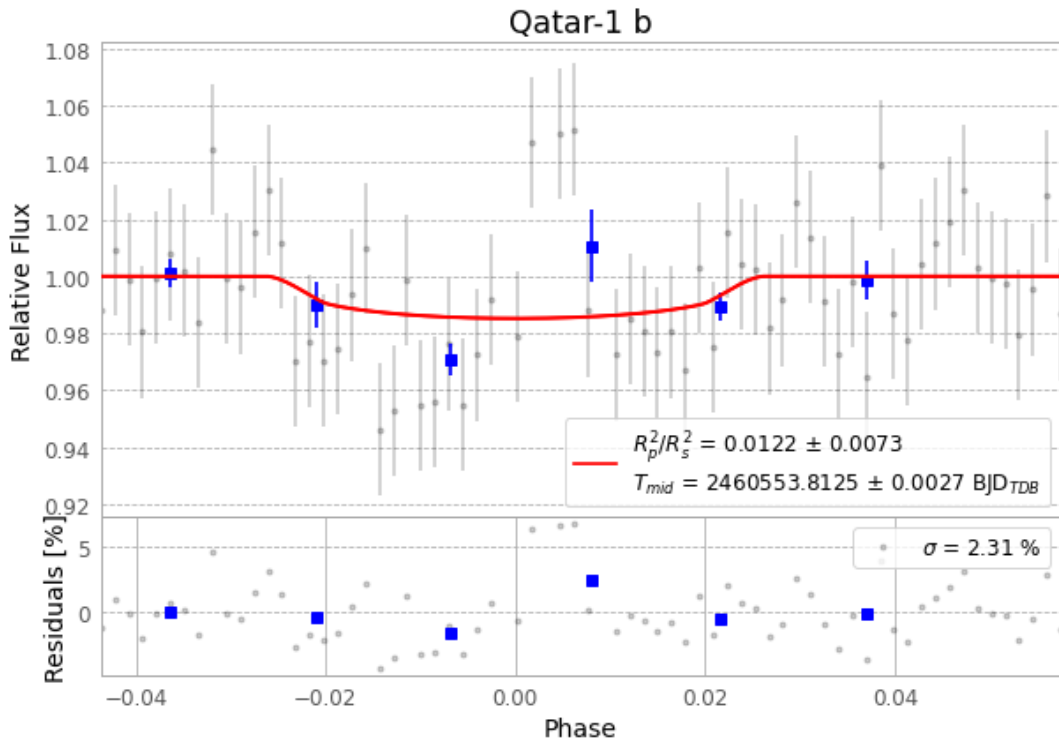


Figure 1 — FinalLightCurve_Qatar-1 b_2024-08-30.png (SHA-256 2cffa53cb6852037...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5a531a975d9c7fbc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

70 FITS frames (46 MB), Qatar-1240831055715.FITS → Qatar-1240831092415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-240831.log (f03001a51359...), FinalLightCurve_Qatar-1 b_2024-08-30.png (2cffa53cb685...), FinalLightCurve_Qatar-1 b_2024-08-30.csv (361a13160e23...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 11:14Z.